

Coniferous forest

Conifers are the world's toughest trees.

Their small, needle-shaped leaves can withstand extreme cold and are impervious to strong sunshine and wind, and their relatively narrow, upright habit enables them to grow closely together to form dense, sheltered forest. As a result, conifers thrive where few broad-leaved trees can survive, such as the far north and in mountain ranges. They also flourish in places that have very heavy rainfall. In such areas they form temperate rain forest, home to some of the largest trees in the world.

Boreal forest

Named after Boreas, the Greek god of the north wind, boreal forest, or taiga, is the largest continuous expanse of forest on earth. It covers about 6 million square miles (15 million square km) and stretches in an almost unbroken belt across the far north, often reaching deep into the Arctic. In some places, the belt is over 1,000 miles (1,600 km) wide. Across the boreal forest belt as a whole, winter temperatures routinely drop below -13°F (-25°C), but in some of the coldest regions, such as northeast Siberia, temperatures can fall below -49°F (-45°C). Summers in boreal forest are brief but can be warm.

Compared with the types of forest that occur at lower latitudes, boreal forest has only a handful of tree species and therefore provides only a

limited variety of food for herbivores. Plant diversity is also restricted both by the amount of light that can reach the forest floor and by the high acidity of pine needles.

Even in summer, the interior of the forest is often dark, with a thick layer of dead needles carpeting the floor. Fungi thrive in these conditions, but the

only forest-floor plants that live here are the ones that can tolerate low light levels and acidic soil conditions.

Other than insects, few animals can digest conifer leaves or wood, so most plant eaters concentrate on seeds, buds, and bark, or on berries from low-growing shrubs. However, what this habitat lacks in variety, it more than makes up for in quantity, especially as there is relatively little competition for food. This is one of the reasons why many boreal forest animals, from birds to bears, have far more extensive ranges than species that live in warmer parts of the world.



THE ANIMAL LIFE in coniferous forest provides birds of prey, such as the northern goshawk, with a year-round food supply.



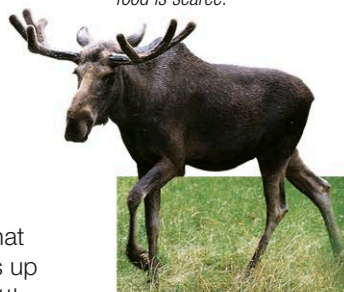
TREE TRUNKS provide the growing larvae of hornets and other wood wasps with protection from the worst of the winter cold.



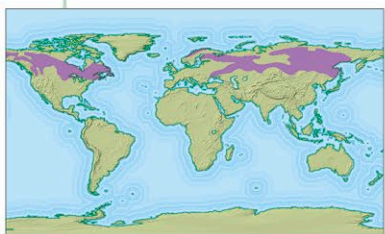
THE BROWN BEAR can climb trees, but it finds most of its food on the ground.



GRAY WOLVES hunt in packs, a strategy that enables them to kill animals larger than themselves. This is particularly important during winter, when food is scarce.



FOR MOOSE and other hoofed mammals, boreal forest offers protection from the cold winds and blizzard conditions of winter.



BOREAL FOREST

stretches across much of the far north. There is no equivalent habitat in the Southern Hemisphere.

CANOPY

UNDERSTORY

GROUND

